

Geographic Anomaly Fraud Detection Policy

SentinelIQ Internal Policy Document - Confidential

1. Purpose and Scope

This policy governs the detection and investigation of geographic anomaly fraud patterns. A geographic anomaly occurs when consecutive transactions on the same account originate from locations that are physically impossible to traverse in the elapsed time between transactions. This indicates either card cloning, credential theft, or account compromise. The policy applies to all transactions processed by SentinelIQ across HDFC, SBI, ICICI, and Axis bank accounts.

2. Definitions

Geographic Anomaly: A condition where two consecutive transactions on the same account occur at locations separated by more than 200 kilometers (haversine distance) with less than 60 minutes elapsed between them. **Haversine Distance:** The great-circle distance between two points on Earth's surface, calculated using latitude and longitude coordinates. **Location Resolution:** The process of mapping a city name or location string to its corresponding (latitude, longitude) coordinate pair using a lookup dictionary of Indian cities. **Impossible Travel:** An informal term for geographic anomaly, referring to the physical impossibility of a legitimate cardholder traversing the distance.

3. Detection Rules

Rule G-1: For each transaction, identify the immediately preceding transaction on the same account_id (by timestamp). Compute the haversine distance between the two locations and the time difference in minutes. If distance > 200 km AND time_difference < 60 minutes, set geo_anomaly_flag to True. **Rule G-2:** If either location cannot be resolved to valid coordinates, the geo_anomaly_flag SHALL be set to False (benefit of doubt). The system returns -1.0 as a sentinel value from the haversine function when resolution fails. **Rule G-3:** Geographic anomalies detected during unusual hours (00:00-04:59) receive compounded risk scoring when combined with the unusual_hour_flag. **Rule G-4:** Geographic anomalies involving cross-city transactions between major metropolitan areas (Mumbai, Delhi, Bangalore, Chennai, Kolkata, Hyderabad) where the distance exceeds 1000 km are automatically escalated to CRITICAL tier regardless of other factors.

4. Haversine Calculation Specification

The haversine formula calculates the shortest distance over the Earth's surface between two points specified by latitude and longitude. Implementation uses Earth radius $R = 6371.0$ km. Formula: $a = \sin(dlat/2)^2 + \cos(lat1) * \cos(lat2) * \sin(dlon/2)^2$ $c = 2 * \arcsin(\sqrt{a})$ distance = $R * c$ Properties: haversine(A, B) == haversine(B, A) (symmetric), haversine(A, A) == 0.0 (identity). The coordinate lookup dictionary includes all major Indian cities and their approximate coordinates. Cities not in the dictionary cannot be resolved, resulting in a -1.0 return value and geo_anomaly_flag set to False.

5. Thresholds and Scoring Impact

A confirmed geographic anomaly contributes 20 points to the base risk score through the active_flag_count mechanism. When combined with velocity patterns, the compound effect is significant: velocity(20) + geo(20) = 40 base points, placing the transaction in at minimum the MED tier. Combined velocity + geo anomalies in unusual hours: base = $3 * 20 = 60$, after night multiplier: $60 * 1.3 = 78$, which places the transaction directly in the HIGH tier for analyst review. The distance threshold of 200 km was selected based on analysis of Indian domestic air travel patterns and typical inter-city commute times. Flights between major Indian cities typically cover 200+ km but require minimum 2-hour door-to-door including airport procedures, making 60-minute windows a reliable impossible-travel indicator.

6. Investigation Procedures

Step 1: Verify location data quality -- confirm both locations resolved correctly in the coordinate lookup. Check for common resolution failures such as abbreviated city names or spelling variations. **Step 2:** Calculate exact distance and time difference. Document both values in the investigation notes. **Step 3:** Review transaction types -- if both transactions

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are DEBIT type at physical merchant locations (not online), the anomaly is stronger evidence of card cloning. Step 4: Check account history for previous geographic anomalies. Repeated anomalies from the same account suggest persistent compromise. Step 5: Cross-reference with velocity patterns -- concurrent velocity burst and geographic anomaly constitutes the combined velocity+geo fraud pattern, one of the 11 distinct patterns in the dataset. Step 6: For cases involving distances > 500 km in < 30 minutes, recommend immediate account suspension pending customer verification.

7. Network Graph Implications

Geographic anomalies often correlate with relay chain patterns in the network graph. When an account shows geographic anomalies while also appearing as a relay node (degree > 5), this strongly indicates the account is being used as a money mule intermediary. The pyvis network visualization on Page 2 renders accounts with geo anomalies using a distinct color coding and displays the distance/time annotation on affected edges. Cycle detection in the directed graph that includes geo-anomalous edges is prioritized for investigation. Merchant nodes receiving transactions from geographically impossible sources are flagged for potential merchant collusion review.

8. Known Limitations and False Positives

The following scenarios may generate false positive geographic anomalies: 1. VPN or proxy transactions where the reported location reflects the proxy server rather than the actual cardholder location. 2. Authorized users on joint accounts transacting from different cities. 3. Online transactions where merchant location differs from cardholder. 4. Travel scenarios where a legitimate user has just arrived in a new city. To mitigate false positives, the confidence level (HIGH/LOW) from the LLM assessment is used to differentiate strong signals from ambiguous ones. LOW confidence geo anomalies in the MED score range (40-69) are escalated to HIGH_QUEUE rather than batch processing, per the routing rules.